

Instructions:

- Solutions to the problem sets are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Particle Velocities (20 pts)

- (a) (5 pts) An electron is accelerated between the plates of a parallel plate capacitor biased to potential V starting from rest at the minus electrode (see Figure 1). Give formulas for the final speed v of the electron in the nonrelativistic approximation and relativistically.
- (b) (5 pts) Using your answer for (a), plug in values $V = 10\text{V}$ and $V = 10\text{ kV}$ for both the nonrelativistic and relativistic formulas. Express answers in m/sec and $\beta = v/c$ units.
- (c) (5 pts) Protons in medical accelerators have kinetic energy $\sim 100\text{ MeV}$. Estimate γ , β , and the speed (in m/sec) of the protons.
- (d) (5 pts) In heavy-ion accelerators ion kinetic energies are typically measured in MeV/nucleon. Show how this approximately fixes the particle axial β . Explain why this is desirable in resonant RF accelerators.

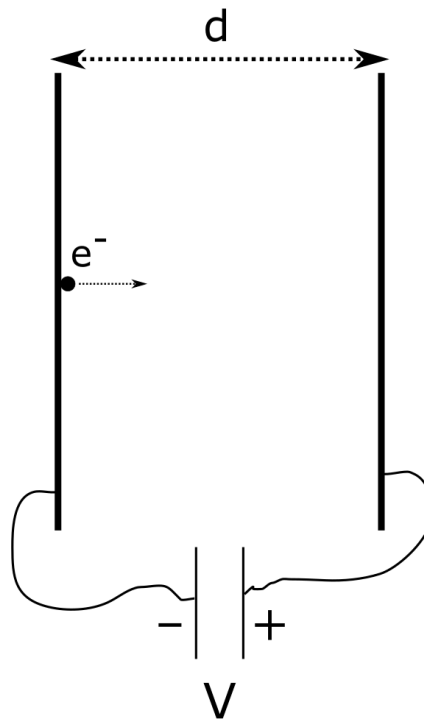


Figure 1: Parallel plate capacitor.

P001 SOLUTION: Particle Velocities 20 pts.

a) 5 pts:

Non-relativistic:

$$T = qV = \frac{1}{2}mv^2 \implies v = \sqrt{\frac{2qV}{m}} = c\sqrt{\frac{2qV}{mc^2}}$$

Relativistic:

$$T = (\gamma - 1)mc^2 = qV, \beta = \frac{1}{\gamma}\sqrt{\gamma^2 - 1}$$

$$\implies \frac{v}{c} = \sqrt{1 - \frac{1}{\left(\frac{qV}{mc^2} + 1\right)^2}}$$

b) 5 pts:

qV	γ	rel. β	rel. v	non-rel. β	non-rel v
10 eV	1.0000196	0.00626	1.86×10^6 m/s	0.0063	1.88×10^6 m/s
10 keV	1.0196	0.195	5.85×10^7 m/s	0.1978	5.93×10^7 m/s

c) 5 pts:

$$(\gamma - 1)mc^2 = 100 \text{ MeV}, mc^2 \approx 938 \text{ MeV}$$

$$\implies \gamma \approx 1.107 \implies \beta \approx 0.429 \implies v \approx 1.286 \times 10^8 \text{ m/s}$$

d) 5 pts:

$$T = (\gamma - 1)mc^2, m \approx Am_u$$

$$\implies \frac{T}{A} \approx (\gamma - 1)m_uc^2 \equiv \text{const.} \cdot f(\beta)$$

Measuring energy this way fixes β in ion machines to help clarify resonance condition needs in RF cavities. Specific ion mass/energy need not be considered. This perspective helps guide ion machine tuning.

Problem 2. RHIC frequency/field (10 pts)

The RHIC collider collides fully stripped gold ions ($A=197$, $Z=79$) at a total energy of $E_{coll}=100$ GeV/nucleon per beam. The circumference of each ring is 3834 m. Assume the mass of a gold ion is 197×0.93113 GeV/ c^2 .

- (a) (5 points) Calculate the revolution frequency of a particle at the injection energy of $E_{inj}=10.5$ GeV/nucleon, and at the storage energy of $E_{coll}=100$ GeV/nucleon. What is the change in revolution frequency for particles accelerated from E_{inj} to E_{coll} ?
- (b) (5 points) If we assume that there are 192 identical dipoles per ring, each of length $L = 10$ m, what is the required dipole field in each at the collision energy of E_{coll} ?

- (a) The revolution frequency can be calculated from the speed and radius

$$T = \frac{1}{f} = \frac{2\pi r}{v} \Rightarrow \frac{v}{2\pi r}$$

The relativistic velocity is $v = \beta c$. We can find β by calculating γ from the energy:

$$U = \gamma mc^2 \Rightarrow \gamma = \frac{U}{mc^2}$$

First, the injection γ is

$$\gamma_{inj} = \frac{(10.5 \text{ GeV/nucleon})(197 \text{ nucleons})}{(197 \times 0.93113 \text{ GeV}/c^2) c^2} = \frac{10.5 \text{ GeV}}{0.93113 \text{ GeV}} = 11.27662$$

This gives

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \frac{1 - \frac{1}{11.27662^2}}{=} 0.99606$$

Substituting $v = \beta c$,

$$f = \frac{\beta c}{2\pi r} = \frac{0.99606c}{3834 \text{ m}} = 77885 \text{ s}^{-1} \Rightarrow \boxed{f_{inj} = 77.885 \text{ kHz}}$$

Following the same process for the storage energy E_{coll} :

$$\gamma_{coll} = \frac{100 \text{ GeV}}{0.93113 \text{ GeV}} = 107.39638$$

$$\beta = \sqrt{1 - \frac{1}{107.39638^2}} = 0.99996$$

$$f = \frac{0.99996c}{3834 \text{ m}} = 78190 \text{ s}^{-1} \Rightarrow \boxed{f_{coll} = 78.190 \text{ kHz}}$$

The change in frequency is

$$\Delta f = f_{coll} - f_{inj} = 78.190 \text{ kHz} - 77.885 \text{ kHz} \Rightarrow \boxed{\Delta f = 0.305 \text{ kHz}}$$

Note: the RHIC accelerating RF cavities actually operate at a harmonic $h = 360$ of the revolution frequency, so they must change frequency from 28.039 MHz to 28.148 MHz.

(b) The magnetic field can be calculated from the rigidity

$$B\rho = \frac{p_0}{|q|}$$

where

$$\theta = \frac{L}{\rho} \implies \rho = \frac{L}{\theta}$$

In total, we need to bend the particle 2π radians with 192 magnets of $L = 10$ m each. Thus

$$\rho = \frac{1920 \text{ m}}{2\pi} = 305.6 \text{ m}$$

The momentum at the collision energy is given by

$$p = \gamma m \beta c$$

and the charge of the ions is $q = Ze$. Plugging in the values from part a, this gives

$$B = \frac{\gamma m \beta c}{Ze\rho} = \frac{(107.39638) \left(197 \times 0.93113 \text{ GeV}/c^2\right) (0.99996c)}{79 (1.6 \times 10^{-19} \text{ C}) (305.6 \text{ m})} \implies \boxed{B = 2.725 \text{ T}}$$

Problem 3. Betatron equations of motion (30 pts)

The betatron equations of motion are

$$x'' - \frac{\rho + x}{\rho^2} = \pm \frac{B_z p_0}{B\rho p} \left(1 + \frac{x}{\rho}\right)^2, \quad z'' = \mp \frac{B_x p_0}{B\rho p} \left(1 + \frac{x}{\rho}\right)^2.$$

Derive these equations for a particle with $p = p_0$ and $q = e$ through the following geometric argument. Let $(\hat{x}, \hat{s}, \hat{z})$ be local polar coordinates inside a dipole (Figure 2). The particle coordinate is

$$\vec{r} = (\rho + x) \hat{x} + z \hat{z},$$

where ρ is the bending radius and

$$\hat{x} = -\rho \frac{d\hat{s}}{ds}.$$

In the case of a planar orbit,

$$\hat{x}' = \frac{1}{\rho} \hat{s},$$

where the prime corresponds to the derivative with respect to s . (See SY Lee section 2.I.1 for more details.) The momentum of the particle is $\vec{p} = \gamma m \dot{\vec{r}}$, where γ is constant in the static magnetic field, and the overdot corresponds to the derivative with respect to time t . Similarly, $d\vec{p}/dt = \gamma m \ddot{\vec{r}}$.

(a) (10 pts) Show that

$$\begin{aligned} \dot{\vec{r}} &= \dot{x} \hat{x} + (\rho + x) \dot{\theta} \hat{s} + \dot{z} \hat{z}, \\ \ddot{\vec{r}} &= \left[\ddot{x} - (\rho + x) \dot{\theta}^2 \right] \hat{x} + \left[2\dot{x}\dot{\theta} + (\rho + x) \ddot{\theta} \right] \hat{s} + \ddot{z} \hat{z}, \end{aligned}$$

where $\theta = s/\rho$ is the angle associated with the reference orbit, i.e. $ds = \rho d\theta$.

- (b) (10 pts) Using $d\vec{p}/dt = e\vec{v} \times \vec{B}$, with $\vec{B} = B_x\hat{x} + B_z\hat{z}$, show that

$$\ddot{x} - (\rho + x)\dot{\theta}^2 = \frac{v_s^2 B_z}{B\rho}, \quad \ddot{z} = -\frac{v_s^2 B_x}{B\rho},$$

where $B\rho = \gamma m v_s / e$ is the momentum rigidity and v_s is the longitudinal velocity.

- (c) (10 pts) Recall that the chain rule for a second derivative is

$$\frac{d^2 y}{dx^2} = \frac{d^2 y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2 u}{dx^2}$$

and assume $d^2 s / dt^2 = 0$. Transform the time coordinate to the longitudinal distance s with $ds = \rho d\theta$, where $d\theta = v_s dt / (\rho + x)$, and show that

$$x'' - \frac{\rho + x}{\rho^2} = \frac{B_z}{B\rho} \left(1 + \frac{x}{\rho} \right)^2, \quad z'' = -\frac{B_x}{B\rho} \left(1 + \frac{x}{\rho} \right)^2.$$

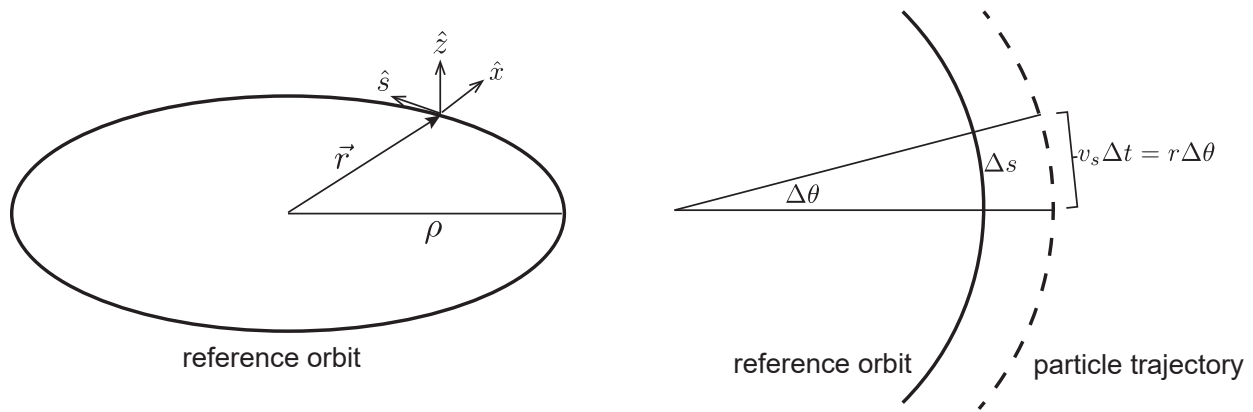


Figure 2: Coordinate system.

- (a) Starting from $\vec{r} = (\rho + x)\hat{x} + z\hat{z}$ and differentiating,

$$\dot{\vec{r}} = \dot{x}\hat{x} + (\rho + x)\dot{\hat{x}} + \dot{z}\hat{z}$$

($\dot{\hat{z}} = 0$). Noting that

$$\hat{x} = -\rho \frac{d\hat{s}}{ds},$$

then

$$\dot{\hat{x}} = \frac{ds}{dt} \frac{d\hat{x}}{ds} = \frac{ds}{dt} \left(\frac{1}{\rho} \hat{s} \right).$$

We can replace ds/dt using $\theta = s/\rho$,

$$\dot{\theta} = \frac{ds}{dt} \frac{d\theta}{ds} = \frac{ds}{dt} \frac{1}{\rho}$$

$$\frac{ds}{dt} = \dot{s} = \rho\dot{\theta},$$

then

$$\boxed{\dot{\vec{r}} = \dot{x}\hat{x} + (\rho + x)\dot{\theta}\hat{s} + \dot{z}\hat{z}} \quad (5 \text{ points})$$

Performing the second derivative,

$$\ddot{\vec{r}} = \ddot{x}\hat{x} + \dot{x}\dot{x} + (\rho + x)\ddot{\theta}\hat{s} + (\rho + x)\dot{\theta}\dot{\hat{s}} + \dot{x}\dot{\theta}\hat{s} + \ddot{z}\hat{z}$$

We can use the relationships we've already found to simplify:

$$\dot{\hat{s}} = \frac{ds}{dt} \frac{d\hat{s}}{ds} = (\rho\dot{\theta}) \left(\frac{-\hat{x}}{\rho} \right) = -\dot{\theta}\hat{x}, \quad \dot{\hat{x}} = \dot{\theta}\hat{s}$$

Substituting, canceling, and rearranging gives

$$\boxed{\ddot{\vec{r}} = \left[\ddot{x} - (\rho + x)\dot{\theta}^2 \right] \hat{x} + \left[2\dot{x}\dot{\theta} + (\rho + x)\ddot{\theta} \right] \hat{s} + \ddot{z}\hat{z}} \quad (5 \text{ points})$$

(b) Recall that

$$\frac{d\vec{p}}{dt} = \gamma m \ddot{\vec{r}} = e\vec{v} \times \vec{B}$$

Taking the cross product $\vec{v} \times \vec{B}$ will give

$$\vec{v} \times \vec{B} = \hat{s}(v_z B_x - v_x B_z) + \hat{x}(v_s B_z) - \hat{z}(v_s B_x)$$

Looking at the $\hat{x}$ and $\hat{z}$ components of the cross product and $\ddot{\vec{r}}$,

$$\ddot{x} - (\rho + x)\dot{\theta}^2 = e \left(\frac{v_s B_z}{\gamma m} \right)$$

$$\ddot{z} = e \left(-\frac{v_s B_x}{\gamma m} \right)$$

Using the substitution $B\rho = \gamma v_s/e$ to rewrite in terms of the rigidity,

$$\boxed{\ddot{x} - (\rho + x)\dot{\theta}^2 = \frac{v_s^2 B_z}{B\rho}} \quad (5 \text{ points})$$

$$\boxed{\ddot{z} = -\frac{v_s^2 B_x}{B\rho}} \quad (5 \text{ points})$$

(c) Using the chain rule as described,

$$\frac{d^2 x}{dt^2} = \frac{d^2 s}{dt^2} \frac{dx}{ds} + \left(\frac{ds}{dt} \right)^2 \frac{d^2 x}{ds^2}, \quad \frac{d^2 s}{dt^2} = 0$$

$$\ddot{x} = \dot{s}^2 x'' = \rho^2 \dot{\theta}^2 x''$$

We also have

$$\dot{\theta} = \frac{v_s}{\rho + x}$$

so

$$\ddot{x} = x'' \frac{\rho^2 v_s^2}{(\rho + x)^2}$$

Substituting,

$$x'' \frac{\rho^2 v_s^2}{(\rho + x)^2} - (\rho + x) \frac{v_s^2}{(\rho + x)^2} = \frac{v_s^2 B_z}{B\rho}$$

We can divide through by the coefficient of the x'' term, rearrange/cancel terms, and arrive at the desired result:

$$\boxed{x'' - \frac{\rho + x}{\rho^2} = \frac{B_z}{B\rho} \left(1 + \frac{x}{\rho}\right)^2} \quad (5 \text{ points})$$

Following the same process for z ,

$$\ddot{z} = \dot{s}^2 z''$$

Substituting in

$$\dot{s} = \rho \dot{\theta} = \frac{\rho}{\rho + x}$$

and simplifying, we get

$$\boxed{z'' = -\frac{B_x}{B\rho} \left(1 + \frac{x}{\rho}\right)^2} \quad (5 \text{ points})$$

,